# THROUGH EVERY POINT OF THE IMAGE OF A FLAT FINITELY PRESENTED MAP THERE IS A LOCALLY QUASI-FINITE FLAT CHART

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 6436-6449 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6451-6458, the complete author proof (6 body lines).

*Editorial note.* Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

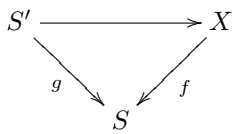
*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The lemma that the Cohen-Macaulay locus of a flat finitely presented morphism is open and meets every fibre in closed points meeting the image condition (same chapter, lemma-flat-finite-presentation-CM-open) and the CM slice lemma producing a locally quasi-finite chart through a CM point (lemma-slice-CM, same chapter); cited inputs with the assembly written in the retained span.

*Source reference.* Original lemma 045U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5981–5984. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0570: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6358–6361. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : X \to S$ be a flat morphism of schemes which is locally of finite presentation. Let $s \in S$ be a point in the image of $f$. Then there exists a commutative diagram*




*where $g : S' \to S$ is flat, locally of finite presentation, locally quasi-finite, and $s \in g(S')$.*

*Proof.* The fibre $X_s$ is not empty by assumption. Hence there exists a closed point $x \in X_s$ where $f$ is Cohen-Macaulay, see Lemma 045U. Apply Lemma 0570 and set $S' = Z$. □

*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.